



[Glossary](#)

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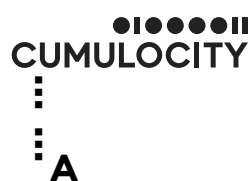
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ADMINISTRATION APPLICATION

The Administration application is one of the default applications of Cumulocity. The Administration application enables administrators to manage their tenants, users, roles and applications. It also allows them to configure various settings for their accounts.

For details see [Platform administration > Standard tenant administration](#).

AGENT

Agents are software components which translate the device-specific interface protocol into a single reference protocol and enable secure bi-directional communication in various network architectures.

ALARM

An alarm represents an [event](#) that requires an intervention. The user or operator of the system must take action to resolve the alarm (like a power outage).

ANALYTICS BUILDER

Analytics Builder is part of the [Cumulocity Streaming Analytics](#) application. It allows you to build analytic models that transform or analyze streaming data in order to generate new data or output events. The models are capable of processing data in real time.

See also [EPL Apps](#).

For details see [Analytics > Streaming analytics > Analytics Builder](#).

APPLICATION

The Cumulocity User Interface (UI) is based on applications. By default, the Cumulocity UI consists of three main default applications: [Administration](#), [Cockpit](#) and [Device Management](#).

Cumulocity is designed to accommodate arbitrary vertical IoT applications in addition to its generic functionality. These applications can have two forms:

- web-based user interface applications (web applications)
- server-side business logic through microservices (microservices)

For details see [Application enablement & solutions > Introduction > Application enablement](#).

APPLICATION SWITCHER

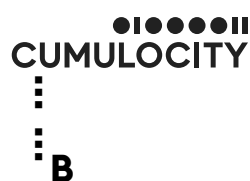
The application switcher is a UI component which shows all Cumulocity applications the user has access to and allows switching between these applications.

ASSET

Assets represent business objects in general like buildings, machines, production units or cars. In Cumulocity, assets are organized in hierarchies which are composed of two types of objects:

- Groups: Objects which group single devices or other groups.
- Devices: Devices which can be linked into the asset hierarchy, after being connected to Cumulocity.

For details see [Application enablement & solutions > Cockpit > Managing assets](#).



BLOCKS

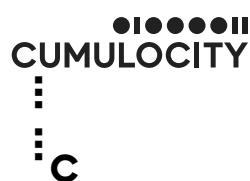
Used in the context of [Analytics Builder](#).

Blocks are the basic processing units of the [model](#). Each block has some predefined functionality and processes data accordingly. A block can have a set of parameters and a set of input ports and output ports.

BUILD ARTIFACT

In the release notes for yearly releases, each change entry includes information on the related build artifact. The term “build artifact” refers to the technical software component in the build pipeline, for example, “c8y-ui”.

See also [Product area](#) and [Component](#).



COCKPIT APPLICATION

The Cockpit application is one of the default applications of Cumulocity. It provides options to manage and monitor IoT assets and data from a business perspective, like managing assets, visualizing data, working with dashboards and managing reports.

For details see [Application enablement & solutions > Cockpit](#).

COMPONENT

In the change logs (or release notes for yearly releases respectively) changes can be filtered for various parameters one of which is the component. Components reflect functional subcategories from a user perspective, for example, "Administration" or "Authentication".

See also [Product area](#).

CONTINUOUS DEPLOYMENT MODEL

In the Cumulocity Continuous Deployment (CD) model, the different components of the Cumulocity platform are updated on a continuous basis. This means a frequent deployment of small incremental improvements. Continuous updates provide customers with a steady stream of enhancements, bug fixes, and new functionalities as they become available.

See also [Private Preview](#), [Public Preview](#) and [General Availability](#).

CUMULOCITY CORE

Cumulocity is the foundation of the Cumulocity platform. It comprises all major components such as the default [applications](#) (Administration, Cockpit and Device Management), the [Cumulocity operational store](#), [microservices](#), [REST API](#), [MQTT API](#), and [smart rules](#).

CUMULOCITY DATAHUB

Cumulocity Datahub is a Cumulocity application for offloading data from the operational store of Cumulocity to a data lake and querying the data lake contents.

For details see [Analytics > DataHub](#).

CUMULOCITY EDGE

Cumulocity Edge is the onsite solution of Cumulocity intended to run as a local software application on industrial PC's or local servers.

For details see [Edge](#).

CUMULOCITY OPERATIONAL STORE

Internal datastore of Cumulocity where all data (such as [alarms](#), [events](#), [inventory](#), and [measurements](#)) are stored in so-called base collections.

CUMULOCITY SENSOR APP

The Cumulocity Sensor App is a free smartphone application available for iOS and Android smartphones. The app is designed to collect measurements from your smartphone, nearby Bluetooth device sensors, and vehicle On-board Debug (OBD) sensors, and send them to the Cumulocity platform. It has a straightforward registration workflow to get you up and running quickly and provides an easy way to get data into Cumulocity.

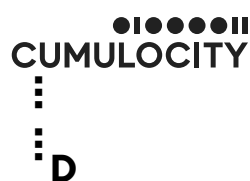
For details see [Cumulocity Sensor App](#).

CUMULOCITY STREAMING ANALYTICS

Using the Streaming Analytics application, you can add your own logic to your IoT solution for immediate processing of incoming data from devices or other data sources. These user-defined operations can, for example, alert applications of new incoming data, create new operations based on the received data (such as sending an alarm when a threshold for a sensor is exceeded), or trigger operations on devices. The operation logic is based on [Apama's Event Processing Language \(EPL\)](#).

See also [Analytics Builder](#) and [EPL Apps](#).

For details see [Analytics > Streaming Analytics](#).



DATA BROKER

The data broker functionality is an optional feature in Enterprise tenants which lets you share data selectively with other tenants.

For details see [Platform administration > Using the data broker](#).

DATA EXPLORER

Part of the [Cockpit application](#). The data explorer visualizes all data points (that is, measurements or sensor data) of either a particular asset or of all assets.

For details see [Application enablement & solutions > Cockpit > Data explorer](#).

DATA LAKE

Used in the context of [DataHub](#).

A data lake serves as a storage container for offloaded data either on the basis of ADLS Gen2/Azure Storage (Azure), S3 (Amazon), NAS.

DATA POINT LIBRARY

The Data point library provides a collection of data points with default values for data point properties which serve as templates that can be applied easily to your data points from different devices.

For details see [Application enablement & solutions > Cockpit > Data point library](#).

DEVICE

Devices are a special type of [asset](#). They can constitute a hierarchy of devices. In the Cumulocity [inventory](#) their [digital twins](#) are represented as [managed objects](#).

DEVICE MANAGEMENT APPLICATION

The Device Management application is one of the default applications of Cumulocity. The Device Management application provides functionalities for connecting, managing and monitoring devices and allows to control and troubleshoot devices remotely.

For details see [Device management & connectivity > Device Management application](#).

DEVICE PROTOCOL

A device protocol defines how data from a specific device type should be transferred into the Cumulocity platform. It contains protocol-specific information about where the data is obtained from, how to transform it, and how to store it in Cumulocity. The definition of the device protocol differs based on the underlying protocol used by the device (for example, Modbus, LoRa, OPC UA or LWM2M).

For details see [Device management & connectivity > Device integration](#).

DIGITAL TWIN

A digital twin is a digital representation of physical or virtual assets. Devices, sensors and tools are examples of physical assets, virtual assets can for example be processes or rules. In Cumulocity they are represented as [managed objects](#).

DOCUMENT

Cumulocity stores all operational data - such as managed objects, alarms and measurements - as documents. A document is a versatile JSON structure with a unique identifier, allowing it to be referenced in URLs of web applications or REST APIs. This provides a high degree of flexibility for modeling your data on top of the Cumulocity [domain model](#) and the Cumulocity [service quotas](#).



ENTERPRISE TENANT

A tenant type in the [Cumulocity tenant hierarchy](#).

Enterprise tenants offer additional administrative functionality compared to a Standard tenant, the major difference being multi-tenancy. Using an Enterprise tenant, you can create and manage subtenants, manage the subscribed applications/features of the subtenants, and invoice subtenants based on usage statistics. Moreover, Enterprise tenants offer individual customization features, such as Branding for the creation of an individual look & feel.

See also [Standard tenant](#) and [Management tenant](#).

EDGE SERVER

Edge Server is an informal term sometimes used to refer to Cumulocity Edge, a single-server variant of the Cumulocity platform, designed to run in factories on industrial PCs or local servers, that is, in the same site (“onsite”) in which the IoT assets are located. For details, see the [Cumulocity Edge documentation](#).

See also [Thick Edge](#)

EVENT PROCESSING LANGUAGE (EPL)

On top of Cumulocity you can use the Apama streaming analytics engine to define business operations for real-time processing. The operation logic is implemented in [Apama’s Event Processing Language \(EPL\)](#). EPL covers statements, which are organized into actions and monitors. EPL files can be edited directly from within Cumulocity using the [Cumulocity Streaming Analytics](#) application. Alternatively, you can develop them locally using the Apama Extension for VS Code. You can deploy your EPL files as Apama applications to Cumulocity.

For more details see the [Apama documentation](#).

EPL APPS

EPL Apps is part of the [Cumulocity Streaming Analytics](#) application. It allows you to develop EPL apps (that is, single *.mon files) directly within Cumulocity, written in Apama EPL. You can also import existing *.mon files as EPL apps into Cumulocity. When you activate an EPL app from the Streaming Analytics application, you deploy it to Cumulocity.

See also [Event Processing Language \(EPL\)](#) and [Analytics Builder](#).

For more details see [Analytics > Streaming Analytics > EPL Apps](#).

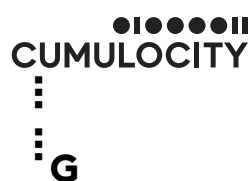
EVENT

In the context of Cumulocity:

Events contain real-time information from the sensor network, such as the triggering of a door sensor. Events can also be [alarms](#). In addition, security-related events are shown as audit logs.

In the context of Apama:

Conceptually, an event is an occurrence of a particular item of interest at a specific time. Apama events are used for all interactions with Cumulocity, such as listening for and creating device measurements, alarms and (Cumulocity) events.



GENERAL AVAILABILITY (GA)

In the Continuous deployment model, General Availability (GA) refers to the availability of a feature to all customers.

See also [Private Preview](#) and [Public Preview](#).

In the Yearly release model, General Availability (GA) refers to the general availability of a [Yearly release](#), typically on the 31st of March of each year.

GLOBAL ROLES

Role type in the permission concept of Cumulocity. Global roles contain permissions that apply to all data within a tenant.

See also [Inventory roles](#) and [Permissions](#).

For details see [Getting started > Technical concepts > Security aspects > Access control](#) and [Platform administration > Standard tenant administration > Managing permissions > Global roles](#).



INVENTORY

The inventory is a collection of all stored [managed objects](#) in a tenant.

INVENTORY ROLES

Role type in the permission concept of Cumulocity. Inventory roles contain permissions that apply to [managed objects](#).

See also [Global roles](#) and [Permissions](#).

For details see [Platform administration > Standard tenant administration > Managing permissions > Inventory roles](#).



LWM2M

Lightweight M2M (LWM2M) is a traffic and resource-optimized protocol to remotely manage IoT devices. Cumulocity provides support for easily connecting any LWM2M device to the platform.

For details see [Device management & connectivity > Device integration > LWM2M](#).



MAINTENANCE RELEASE

Cumulocity provides Maintenance releases for supported Cumulocity [Yearly releases](#). A Maintenance release contains fixes for a Yearly release but no new features.

MANAGED OBJECTS

In Cumulocity, all [assets](#) are represented as managed objects. Cumulocity comes with a predefined set of types, for example [devices](#), groups, and [agents](#).

Hierarchical structures of managed objects allow for the implementation of complex structures like a factory with many machines which consist of devices and sensors which can report [events](#), [alarms](#), and [measurements](#).

MANAGEMENT TENANT

A tenant type in the [Cumulocity tenant hierarchy](#).

The Management tenant builds the highest level of the tenant hierarchy. Every Cumulocity deployment is delivered with a Management tenant. The Management tenant is used to administer all tenants within the same deployment on platform level and thus provides full control of the platform.

See also [Standard tenant](#) and [Enterprise tenant](#).

MEASUREMENT

A measurement contains one or multiple numerical values taken at a single point in time for a single asset. A sequence of measurements forms a time series.

MICROSERVICES

Microservices are server-side applications. Microservices can be used to develop for example the following functionality on top of Cumulocity:

- Integrations
- Batch analytics
- Decoder
- Backend applications

Microservices are deployed as Container images to Cumulocity, and follow specific conventions. They typically provide one REST API, which is available under /service/ and access Cumulocity using this REST API.

For details see [Platform administration > Standard tenant administration > Managing the ecosystem > Managing microservices](#) and [Application enablement & solutions > Microservice SDK](#).

MODEL

Used in the context of [Analytics Builder](#).

A model is a container which can have a network of [blocks](#) connected to each other with wires. The behavior of a block inside a model does not depend on other blocks. There can be multiple instances of the same block in a model where each instance may behave differently, depending on the configurable parameters or the inputs connected to the block.

MQTT

Cumulocity supports MQTT, an OASIS standard messaging protocol, for device integration.

For details see [Device management & connectivity > Device integration > MQTT](#).

MULTI-TENANCY

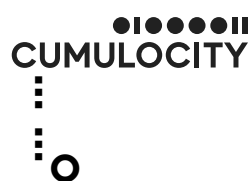
With the Enterprise tenant concept, Cumulocity supports full multi-tenancy. All data related to a tenant is stored in a dedicated database. This includes user data, inventory, events, measurements, operations and alarms.

For details see [Getting started > Technical concepts > Tenant hierarchy > Multi-tenancy](#).



NAVIGATOR

The navigator is an element in the UI of the Cumulocity platform. It is located at the left of the UI. The navigator provides a list of menu items which lead you to the various pages of the application you are currently using.



OPC UA

OPC Unified Architecture (OPC UA) is a standard to enable the communication between industrial devices. OPC UA is designed to work across technology boundaries (cross-platform).

One component of the OPC UA integration in Cumulocity is the OPC UA device gateway. The OPC UA device gateway is a stand-alone Java program that communicates with OPC UA server(s) and the Cumulocity platform. It stores data into the Cumulocity database via REST. The gateway must be registered as Cumulocity device.

For details see [Device management & connectivity > Device integration > OPC UA](#).

OPERATIONS

Operations are cloud-to-device messages so that devices can be remotely controlled and managed by Cumulocity. Typical operations are installing a new software, switching a relay in a power meter or sending a credit to a vending machine.

For details see [Getting started > Technical concepts > Cumulocity's domain model > Operations](#).



PERMISSIONS

The ability to execute certain functionality in the Cumulocity platform is based on permissions. Permissions define explicitly what functionality can be executed by a user. Cumulocity distinguishes read permissions and administration permissions. Read permissions enable users to read data. Administration permissions enable users to create, update and delete data.

To manage permissions more easily, they are grouped into roles. Every user can be associated with a number of roles, adding up permissions of the user.

See also [Global roles](#) and [Inventory roles](#).

PRIVATE PREVIEW MODE

In the [Continuous Deployment](#) model, the Private Preview mode refers to a feature which is only made available to selected customers.

See also [General Availability](#) and [Public Preview](#).

PRODUCT AREA

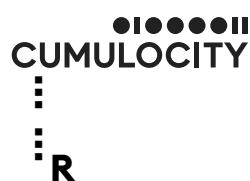
In the change logs (or release notes for yearly releases respectively) changes can be filtered for various parameters one of which is the product area. The product area is a high-level functional area, for example, "Device management & connectivity".

See also [Components](#).

PUBLIC PREVIEW MODE

In the [Continuous Deployment](#) model, the Public Preview mode refers to a feature for which customers can opt-in to have the feature activated.

See also [General Availability](#) and [Private Preview](#).



REAL-TIME PROCESSING

Cumulocity allows developers and power users to run real-time IoT business logic inside Cumulocity based on a high-level real-time processing language, that is, Apama's Event Processing Language (EPL).

For details see [Analytics > Streaming Analytics](#).

REST

In Cumulocity, REST (Representational State Transfer) is used for all external communication with IoT devices, other web applications or back-office IT systems.

REST API

The Cumulocity REST API is an interface that allows for communication between the Cumulocity platform and other systems via HTTP and REST. It allows the user to perform operations and process data by issuing a simple request using any HTTP client such as a web browser.

For details see the [Cumulocity OpenAPI Specification](#).



404

PAGE NOT FOUND

The page you are looking for does not exist or has been moved.

Click [here](#) to go back to the documentation home page.



SMART RULES

Cumulocity includes a rule engine to analyze data in realtime and to perform actions based on data. To easily create rules, the Cockpit application includes a “smart rules” builder which allows you to create rules from templates (so-called smart rule templates).

For details see [Application enablement & solutions > Cockpit > Smart rules](#).

STANDARD TENANT

A tenant type in the [Cumulocity tenant hierarchy](#).

At the bottom of the tenant hierarchy you can find single tenants which are represented by the concept of Standard tenant. A Standard tenant offers most of the device management and monitoring functionality of the Cumulocity platform, but has certain limitations when it comes to administrative aspects.

See also [Enterprise tenant](#) and [Management tenant](#).

SUBSCRIPTION

The application concept of Cumulocity includes a basic application marketplace. Tenants can be subscribed to applications which have been deployed by their superior tenant (Management tenant or Enterprise tenant). Granting access to subtenants and subscribing to applications is done in the Administration application.

For details see [Platform administration > Enterprise tenant administration > Managing tenants > Subscribing applications](#) and [Platform administration > Standard tenant administration > Managing the ecosystem > Managing applications](#).



TENANT

Tenants are physically separated data spaces with a separate URL, which has a specific set of users, a separate application management and no data sharing by default. Users in a single tenant share the same URL and the same data space.

See also [Tenant hierarchy](#).

TENANT DOMAIN

A key feature of the Enterprise tenant is the ability to operate the Cumulocity platform using a custom domain name. This means that you can configure the platform to serve you and your customers using a host name of choice.

For details see [Platform administration > Enterprise tenant administration > Customizing your platform](#).

TENANT HIERARCHY

The Cumulocity tenant concept builds a 3-level hierarchy, including the following levels from bottom to top:

- [Standard tenant](#)
- [Enterprise tenant](#)
- [Management tenant](#)

These three levels differ in their scope, particularly with regards to administration.

See also [Tenant hierarchy](#).

TENANT ID

The unique ID of a tenant or subtenant. When a tenant is created, it gets an auto-generated ID, which cannot be changed. The tenant ID is shown in the user dropdown menu in the UI.

THICK EDGE

Thick Edge is an informal term sometimes used to refer to Cumulocity Edge, a single-server variant of the Cumulocity platform, designed to run in factories on industrial PCs or local servers, that is, in the same site ("onsite") in which the IoT assets are located. For details, see the [Cumulocity Edge documentation](#).

See also [Edge Server](#)

THIN EDGE

[Thin-edge.io](#) is an open-source and cloud-agnostic IoT framework designed for lightweight IoT devices. It offers simple and secure device connectivity, freedom of the cloud platform, for example Cumulocity,

and freedom of the programming language.



WEB APPLICATIONS

A Cumulocity web application can be:

- a user interface application built on any web framework of your choice or
- a user interface application built using the Cumulocity user interface framework as a set of user interface plugins.

All subscribed web applications are hosted by Cumulocity and the application will be made available through a URL like this: `.cumulocity.com/apps/`.

See also [Platform administration > Standard tenant administration > Managing the ecosystem](#).

WEB SDK

The Cumulocity Web SDK enables you to develop web applications that can be:

- deployed to the platform,
- communicate authenticated with our API,
- apply default or
- branded UI components to your custom application.

For details see the [Application enablement & solutions > Web SDK](#).

WIDGETS

Widgets are useful to track information, for example on alarms, assets or applications, or provide maps, quick links and more in dashboards or reports. Cumulocity provides preset widget types for various purposes.

For details see [Application enablement & solutions > Cockpit > Widgets collection](#) and [Application enablement & solutions > Cockpit > Using widgets in dashboards and reports](#).

WIRES

Used in the context of [Analytics Builder](#).

Wires are used to connect two or more [blocks](#) with each other. They are used for all data transfer between the output port of one block and the input port of another block.



YEARLY RELEASE

Yearly releases are provided in a yearly cycle and named after the year, for example, Cumulocity 2024 Release. The [General Availability \(GA\)](#) for a Yearly release is scheduled for the 31st of March of the respective year. Yearly releases are only available to customers with a dedicated Cumulocity instance. For customers self-hosting Cumulocity, Yearly releases are made available via the [Cumulocity Download Center](#). Yearly releases include new features, improvements and fixes.